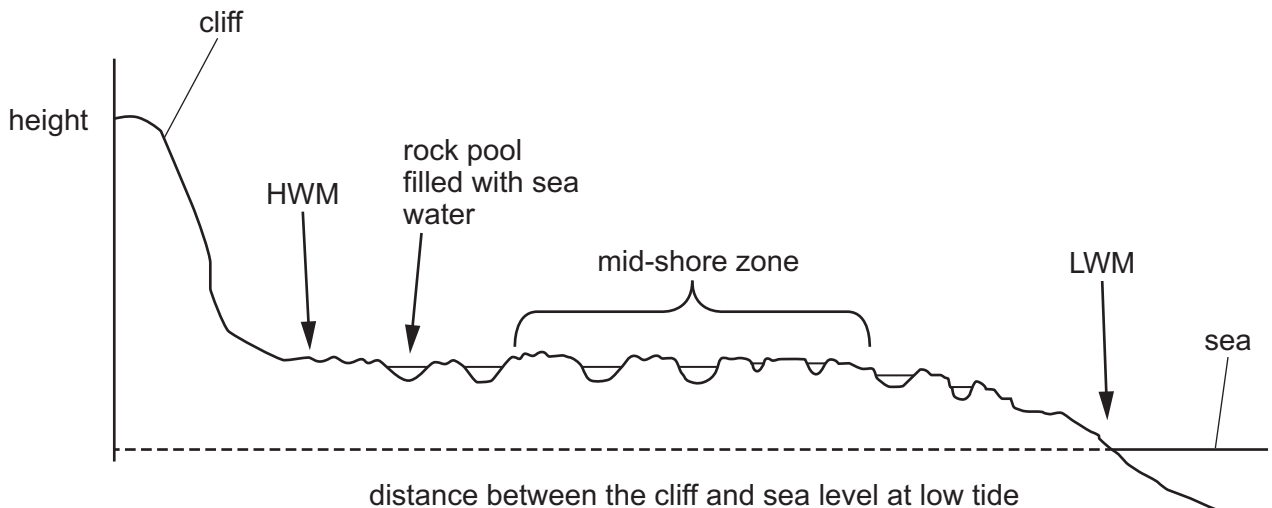


- 1 Rocky shore ecosystems are found in some coastal regions. These ecosystems are inhabited by many different species of algae and animals. Twice each day the sea level rises to the high tide level and decreases to the low tide level in a regular pattern. The organisms in these ecosystems are exposed to very different conditions during the course of each day.

Algae are photosynthetic protocists. Multicellular algae are found in many marine ecosystems.

Rock pools are formed in depressions in the shore as the water level lowers when the tide goes out.

Fig. 1.1 is a profile of a rocky shore ecosystem showing the positions of some rock pools. The mid-shore zone lies between the lower-shore zone and the upper-shore zone.



HWM = high-water mark – the highest point on the shore that is covered by sea water at high tide

LWM = low-water mark – the lowest point on the shore that is exposed to the air at low tide

Fig. 1.1

Fig. 1.2 shows a large rock pool on the mid-shore zone surrounded by smaller rock pools at low tide.



Fig. 1.2

A student investigated the distribution and abundance of large, multicellular algae on a rocky shore in North Wales in the UK at low tide on one day in October 2011.

The student completed a preliminary survey of the shore at low tide and made some observations.

1. There were several different species of algae growing in the mid-shore zone.
2. Some algae were exposed to the air at low tide, but most grew in rock pools.
3. The distribution and abundance of the algae differed on the shore between the low-water mark (LWM) and the high-water mark (HWM).
4. Large rock pools appeared to have more species of algae than small rock pools.
5. Large rock pools were deeper than small rock pools.

The student decided to investigate the relationship between the depth of rock pools and the species diversity of the algae in the rock pools in the mid-shore zone.

The student selected 32 rock pools at random. Approximately half the rock pools were small and half were large. The depth of each rock pool at its deepest point was measured and the number of different species of algae in each rock pool was counted.

(a) State **two** variables that have been standardised in the investigation.

- 1
-
- 2
-

[2]

(b) The results of the investigation are shown in Fig. 1.3.

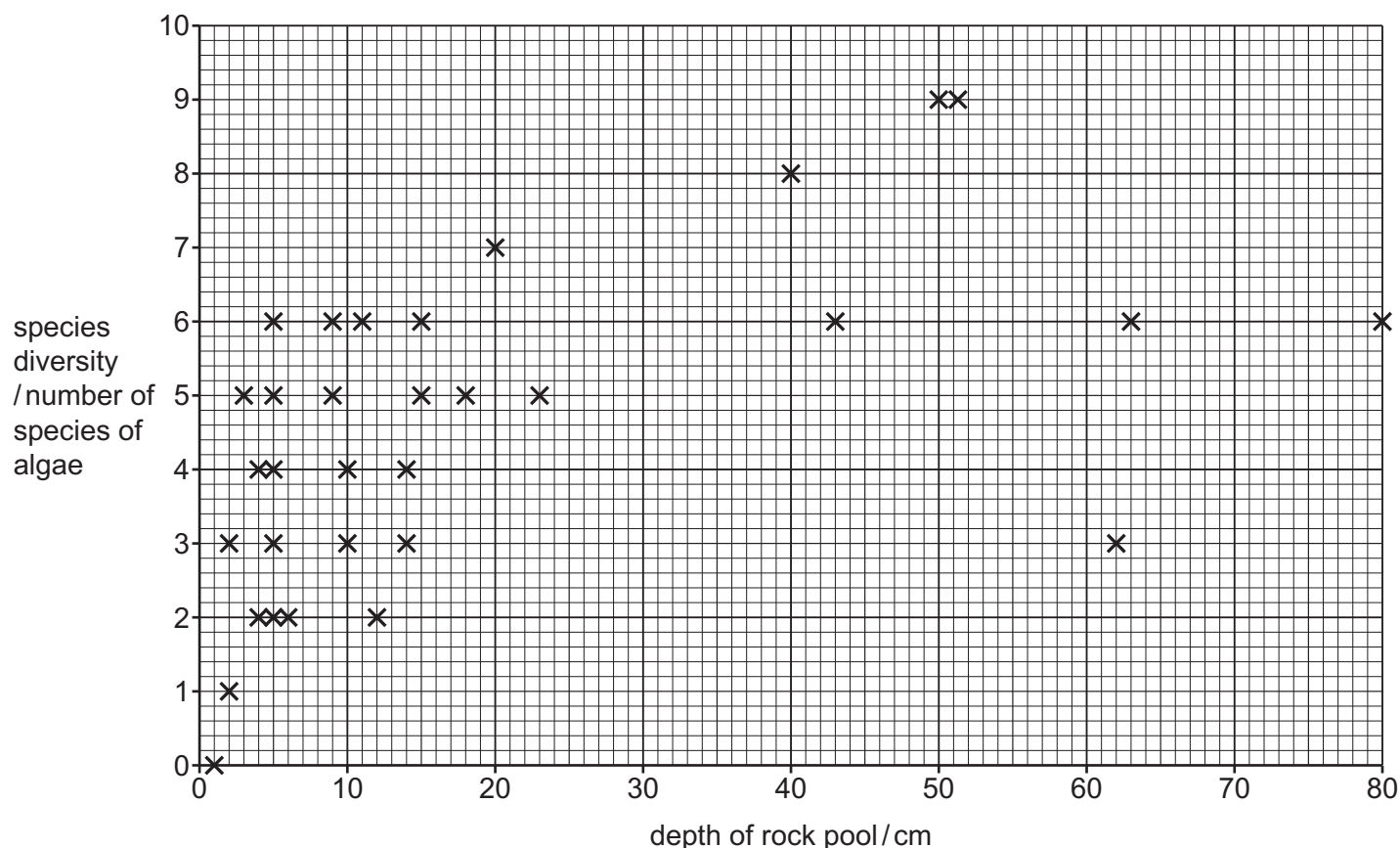


Fig. 1.3

The student used the Spearman's rank correlation to analyse the data in Fig. 1.3.

(i) State a null hypothesis for this investigation.

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 [1]

(ii) The student calculated the Spearman's rank correlation coefficient (r_s) as 0.628.

Table 1.1 shows some of the critical values for the Spearman's rank correlation test.

Table 1.1

number of pairs of measurements	critical values	
	$p = 0.05$ (5%)	$p = 0.01$ (1%)
30	0.362	0.467
31	0.356	0.459
32	0.350	0.452
33	0.345	0.446
34	0.340	0.439

Discuss, with reference to Table 1.1, the conclusions that can be made from the analysis of the data collected by the student.

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..... [3]

(iii) State **one** reason why Pearson's linear correlation coefficient is **not** appropriate for analysing the data shown in Fig. 1.3.

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..... [1]

(c) One important feature of scientific reports is that they provide enough information for other researchers to repeat each investigation.

Explain why it is difficult for researchers to repeat this investigation of species diversity in rock pools from the information provided.

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..... [2]

- (d) The student noticed that part of the rocky shore had **no** rock pools. The student observed that the species of algae were not distributed equally on this part of the rocky shore from the LWM to the HWM.

Describe an investigation to find out how the distribution **and** abundance of the different species of algae varies on a rocky shore with **no** rock pools from the LWM to the HWM.

Your method should be set out in a logical order and be detailed enough for another person to follow.

This image shows a single page of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

[7]

[Total: 16]